GOVERNMENT TECHNICAL INSTITUTE (POL)

Department of Information Technology

TCP/IP
JOB SHEET

2025- Academic YearA.G.T.I

Second year (SemII)

Submitted by
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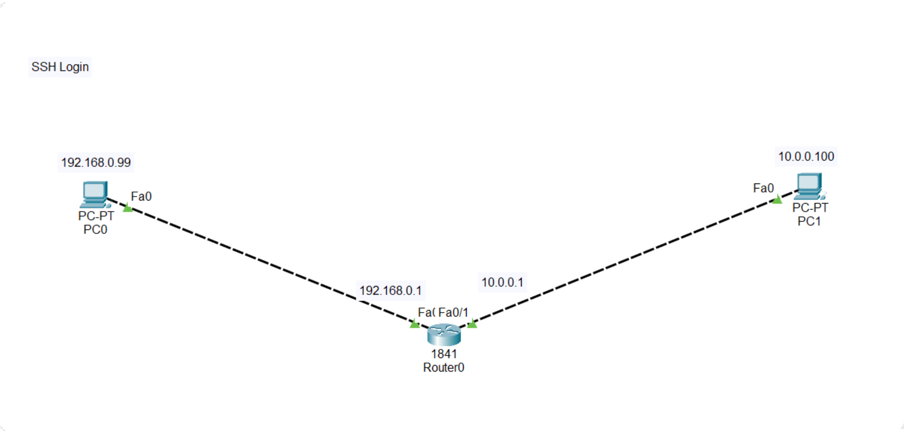
Date: 18.9.2025

Jobsheet-1

SSH Login



For R1:

- **Router>en**
- **Router#conf t**
- **Router(config)#host R1**
- **R1(config)#ip domain-name pyinoolwin.local**
- **R1(config)#crypto key generate rsa**
- **How many bits in the modulus[512]:2048**

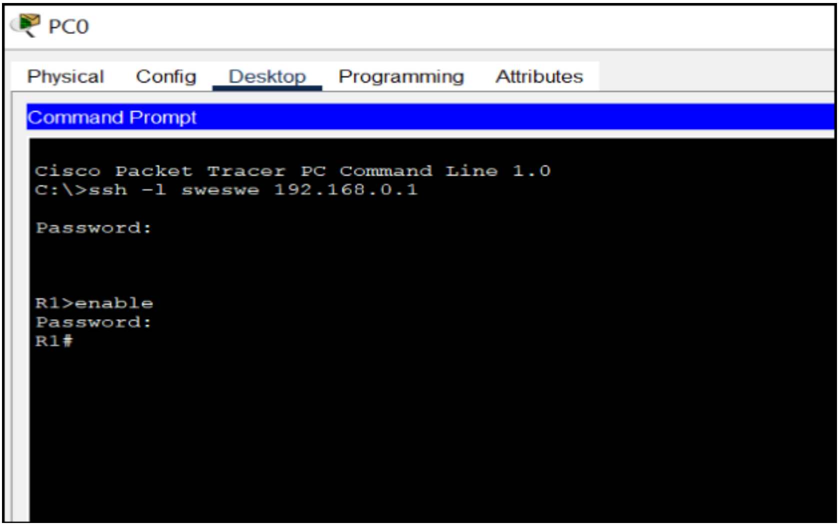
- **R1(config)#ip ssh version 2**
- **R1(config)#line vty 0 4**
- **R1(config-line)#transport input ssh**
- **R1(config-line)#exit**

- **R1(config)#username sweswe password networking**
- **R1(config)#enable secret poltechnologies**
- **R1(config)#line vty 0 4**
- **R1(config-line)#login local**

```
191K bytes of NVRAM.
63488K bytes of ATA CompactFlash (Read/Write)
Cisco IOS Software, 1841 Software (C1841-ADVIPSERVICESK9-M), Version 12.4(15)T1, RELEASE
SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 04:52 by pt_team

Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up


Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#host R1
R1(config)#ip domain-name pyinoolwin.local
R1(config)#crypto key generate rsa
The name for the keys will be: R1.pyinoolwin.local
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#ip ssh version 2
*Mar 1 0:2:17.346: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 04
R1(config-line)#transport input ssh
R1(config-line)#exit
R1(config)#username student password networking
R1(config)#enable secret poltechnologies
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#%IP-4-DUPADDR: Duplicate address 192.168.0.99 on FastEthernet0/0, sourced by
00D0.5807.E909
%IP-4-DUPADDR: Duplicate address 10.0.0.100 on FastEthernet0/1, sourced by 0001.C983.95D2

R1(config-line)#
```

SSH login in PC1

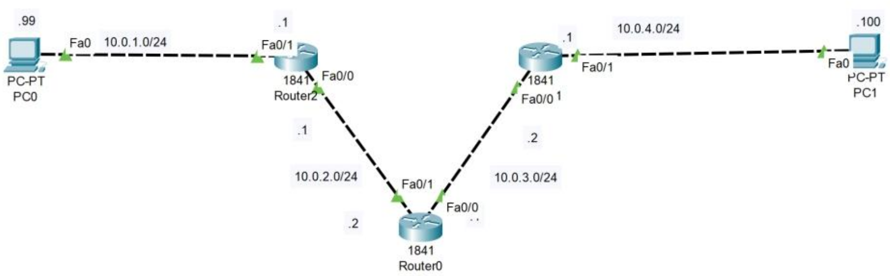
- **ssh -l sweswe 192.168.0.1**
- **Password: networking**
- **R1>enable**



- Password: poltechnologies

Jobsheet-2

Networking-Static Route



Route

Step 1: Take the necessary components and cable them.

Step 2: Configuration in all routers.

<u>For Router0</u>:

- **Router>en**
- **Router#show ip interface brief**
- **Router#conf ter**

- Enter configuration commands, one per line. End with CNTL/Z.
- Router(config)#hostname RB
- RB(config)#enable secret router
- RB(config)#line console 0
- RB(config-line)#password cisco
- RB(config-line)#login
- RB(config-line)#line vty 0 4
- RB(config-line)#password cisco
- RB(config-line)#login
- RB(config-line)#exec-timeout 0 0
- RB(config-line)#int Fa0/1
- RB(config-if)#ip address 10.0.2.2 255.255.255.0
- RB(config-if)#no shutdown
- RB(config-if)#exit
- RB(config)#int Fat0/0
- RB(config-if)#ip address 10.0.3.1 255.255.255.0
- RB(config-if)#no shutdown

For Router1:

- Router>en
- Router#show ip interface brief

- Router#configure terminal
- Enter configuration commands, one per line. End with CNTL/Z.
- Router(config)#hostname RA
- RA(config)#enable secret router
- RA(config)#line console 0
- RA(config-line)#password cisco
- RA(config-line)#login
- RA(config-line)#line vty 0 4
- RA(config-line)#password cisco
- RA(config-line)#login
- RA(config-line)#exec-timeout 0 0
- RA(config-line)#interface Fa0/1
- RA(config-if)#ip address 10.0.1.1 255.255.255.0
- RA(config-if)#no shutdown
- RA(config-if)#
- %LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
- RA(config-if)#interface Fa 0/0
- RA(config-if)#ip address 10.0.2.1 255.255.255.0
- RA(config-if)#no shutdown

For Router2:

- **Router>en**
- **Router#show ip interface brief**
- **Router#conf ter**
- **Enter configuration commands, one per line. End with CNTL/Z.**
- **Router(config)#hostname RC**
- **RC(config)#enable secret router**
- **RC(config)#line console 0**
- **RC(config-line)#password cisco**
- **RC(config-line)#login**
- **RC(config-line)#line vty 0 4**
- **RC(config-line)#password cisco**
- **RC(config-line)#login**
- **RC(config-line)#exec-timeout 0 0**
- **RC(config-line)#int Fa0/0**
- **RC(config-if)#ip address 10.0.3.2 255.255.255.0**
- **RC(config-if)#no shutdown**
- **RC(config-if)#exit**
- **RC(config)#int Fas0/1**
- **RC(config-if)#ip address 10.0.4.1 255.255.255.0**

- RC(config-if)#no shutdown

Step 3:IP Route

<u>For RA:</u>

- RA#sh ip route
- RA(config)#ip route 10.0.3.0 255.255.255.0 10.0.2.2
- RA(config)#ip route 10.0.4.0 255.255.255.0 10.0.2.2
- RA(config)#exit
- RA#sh ip route

<u>For RB:</u>

- RB#sh ip route
- RB(config)#ip route 10.0.1.0 255.255.255.0 10.0.2.1
- RB(config)#ip route 10.0.4.0 255.255.255.0 10.0.3.2
- RB(config)#exit
- RB#sh ip route

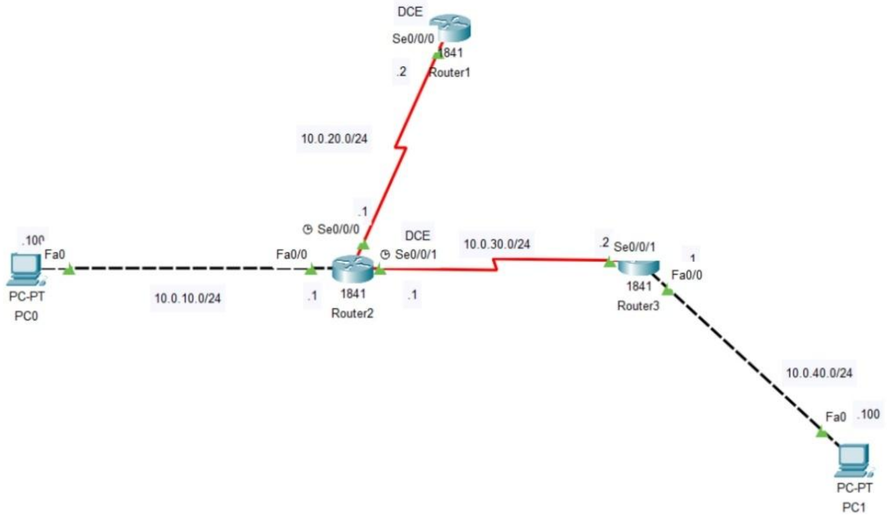
<u>For RC:</u>

- RC#sh ip route
- RC(config)#ip route 10.0.2.0 255.255.255.0 10.0.3.1
- RC(config)#ip route 10.0.1.0 255.255.255.0 10.0.3.1

- **RC#sh ip route**

Jobsheet-3

Dynamic route



<u>Router 1</u>

- Router1>en

- Router1#config t
- Router1(config)#int Se0/0/0
- Router1(config-if)#ip add 10.0.20.2 255.255.255.0
- Router1(config-if)#no shut
- Router1(config-if)#exit
- Router1(config)#exit
- Router1#show ip interface brief
- Router1#config t
- Router1(config)#ip route 0.0.0.0 0.0.0.0 10.0.20.1

Router 2

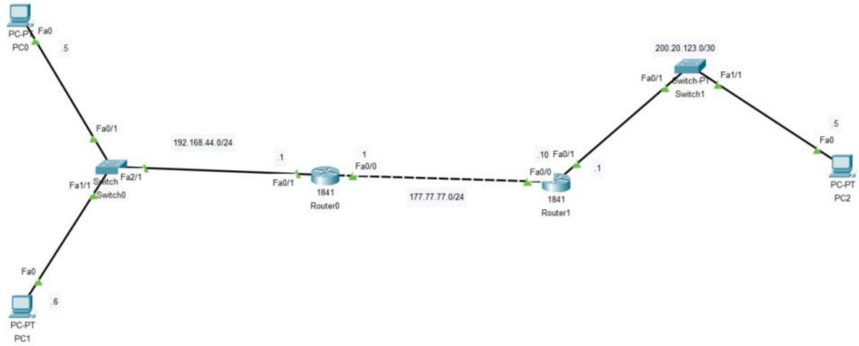
- Router2>en
- Router2#config t
- Router2(config)#int Fa0/0
- Router2(config-if)#ip add 10.0.10.1 255.255.255.0
- Router2(config-if)#no shut
- Router2(config-if)#exit
- Router2(config)#int Se0/0/0
- Router2(config-if)#ip add 10.0.20.1 255.255.255.0
- Router2(config-if)#no shut
- Router2(config-if)#exit
- Router2(config)#int Se0/0/1
- Router2(config-if)#ip add 10.0.30.1 255.255.255.0
- Router2(config-if)#no shut
- Router2(config-if)#exit
- Router2(config)#exit
- Router2#show ip interface brief
- Router2#config t
- Router2(config)#ip route 0.0.0.0 0.0.0.0 10.0.10.2

Router 3

- Router3>en
- Router3#config t
- Router3(config)#int Fa0/0
- Router3(config-if)#ip add 10.0.40.1 255.255.255.0
- Router3(config-if)#no shut
- Router3(config-if)#exit
- Router3(config)#int Se0/0/1
- Router3(config-if)#ip add 10.0.30.2 255.255.255.0
- Router3(config-if)#no shut
- Router3(config-if)#exit
- Router3(config)#exit
- Router3#show ip interface brief
- Router3#config t
- Router3(config)#ip route 0.0.0.0 0.0.0.0 10.0.10.1

Jobsheet-4

NAT CONFIGURATION



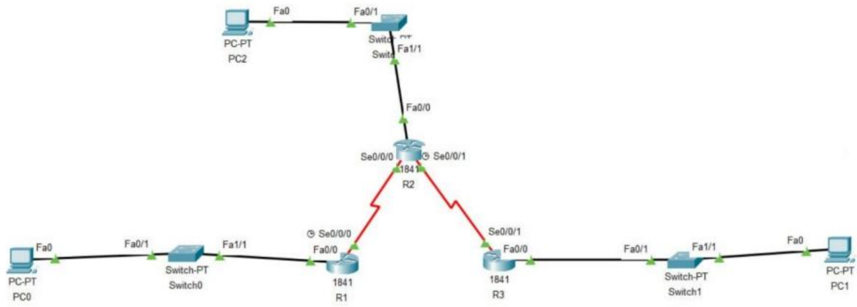
Router 1

- ➢ Router>en
- ➢ Router#conf ter
- ➢ Router(config)#hostname Router1
- ➢ Router1(config)#int FastEthernet0/0
- ➢ Router1(config-if)#ip address 177.77.77.10 255.255.255.0
- Router1(config-if)#no shutdown
- Router1(config-if)#exit
- Router1(config)#int FastEthernet0/1
- Router1(config-if)#ip address 200.20.123.1 255.255.255.0
- Router1(config-if)#no shutdown
- Router1(config-if)#exit
- Router1>en
- Router1#sh ip interface brief
- Router1(config)#ip route 192.168.44.0 255.255.255.0 177.77.77.1
- Router1(config-if)#ip route 192.168.44.0 255.255.255.0 177.77.77.1
- Router1(config)#exit

Router 0

- Router>en
- Router#conf ter
- Router(config)#hostname Router0
- Router0>en
- Router0#sh ip interface brief
- Router0#conf term
- Router0(config)#int FastEthernet0/0
- Router0(config-if)#ip address 192.168.44.1 255.255.255.0
- Router0(config-if)#no shutdown
- Router0(config-if)#exit
- Router0(config)#int FastEthernet0/1
- Router0(config-if)#ip address 177.77.77.1 255.255.255.0
- Router0(config-if)#no shutdownRouter0(config-if)#ip route 200.20.123.0 255.255.255.0 177.77.77.10
- Router0(config)#exit
- Router0(config)#access-list 1 permit 192.168.44.0 0.0.0.255
- Router0(config)#ip nat pool NAT-POOL 177.77.77.1 177.77.77.7 netmask 255.255.255.0
- Router0(config)#ip nat inside source list 1 pool NAT-POOL
- Router0(config)#int FastEthernet0/0
- Router0(config-if)#ip nat inside
- Router0(config-if)#exit
- Router0(config)#int FastEthernet0/1
- Router0(config-if)#ip nat outside

Jobsheet8



For R1

Router>en

Router#config t

Router(config)#hostname R1

R1(config)#int Fa0/0

R1(config-if)#ip add 192.168.1.1 255.255.255.0

R1(config-if)#no shut

R1(config-if)#exit

R1(config)#int Se0/0/0

R1(config-if)#ip add 192.168.2.1 255.255.255.0

R1(config-if)#no shut

R1(config-if)#exit

R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.2.2

R1(config)#router rip

R1(config-router)#network 192.168.1.0

R1(config-router)#network 192.168.2.0

R1(config-router)#end

R1#show ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is 192.168.2.2 to network 0.0.0.0

C 192.168.1.0/24 is directly connected, FastEthernet0/0

C 192.168.2.0/24 is directly connected, Serial0/0/0

R 192.168.3.0/24 [120/1] via 192.168.2.2, 00:00:05, Serial0/0/0

R 192.168.4.0/24 [120/1] via 192.168.2.2, 00:00:05, Serial0/0/0

R 192.168.5.0/24 [120/2] via 192.168.2.2, 00:00:05, Serial0/0/0

S* 0.0.0.0/0 [1/0] via 192.168.2.2

R1#show ip protocols

Routing Protocol is "rip"

Sending updates every 30 seconds, next due in 12 seconds

Invalid after 180 seconds, hold down 180, flushed after 240

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Redistributing: rip

Default version control: send version 1, receive any version

Interface Send Recv Triggered RIP Key-chain

FastEthernet0/0 12 1

Serial0/0/0 12 1

Automatic network summarization is in effect

Maximum path: 4

Routing for Networks:

192.168.1.0

192.168.2.0

Passive Interface(s):

Routing Information Sources:

Gateway Distance Last Update

192.168.2.2 120 00:00:15

Distance: (default is 120)

R1#debug ip rip

RIP protocol debugging is on

R1#RIP: received v1 update from 192.168.2.2 on Serial0/0/0

192.168.3.0 in 1 hops

192.168.4.0 in 1 hops

192.168.5.0 in 2 hops

RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (192.168.1.1)

RIP: build update entries

network 192.168.2.0 metric 1

network 192.168.3.0 metric 2

network 192.168.4.0 metric 2

network 192.168.5.0 metric 3

RIP: sending v1 update to 255.255.255.255 via Serial0/0/0 (192.168.2.1)

RIP: build update entries

network 192.168.1.0 metric 1

RIP: received v1 update from 192.168.2.2 on Serial0/0/0

192.168.3.0 in 1 hops

192.168.4.0 in 1 hops

192.168.5.0 in 2 hops

RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (192.168.1.1)

RIP: build update entries

network 192.168.2.0 metric 1

network 192.168.3.0 metric 2

network 192.168.4.0 metric 2

network 192.168.5.0 metric 3

RIP: sending v1 update to 255.255.255.255 via Serial0/0/0 (192.168.2.1)

RIP: build update entries

network 192.168.1.0 metric 1

RIP: received v1 update from 192.168.2.2 on Serial0/0/0

192.168.3.0 in 1 hops

192.168.4.0 in 1 hops

192.168.5.0 in 2 hops

RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (192.168.1.1)

RIP: build update entries

network 192.168.2.0 metric 1

network 192.168.3.0 metric 2

network 192.168.4.0 metric 2

network 192.168.5.0 metric 3

RIP: sending v1 update to 255.255.255.255 via Serial0/0/0 (192.168.2.1)

RIP: build update entries

network 192.168.1.0 metric 1

RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (192.168.1.1)

RIP: build update entries

network 192.168.2.0 metric 1

network 192.168.3.0 metric 2

network 192.168.4.0 metric 2

network 192.168.5.0 metric 3

RIP: sending v1 update to 255.255.255.255 via Serial0/0/0 (192.168.2.1)

RIP: build update entries

network 192.168.1.0 metric 1

RIP: received v1 update from 192.168.2.2 on Serial0/0/0

192.168.3.0 in 1 hops

192.168.4.0 in 1 hops

192.168.5.0 in 2 hops

RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (192.168.1.1)

RIP: build update entries

network 192.168.2.0 metric 1

network 192.168.3.0 metric 2

network 192.168.4.0 metric 2

network 192.168.5.0 metric 3

For R2

Router>en

Router#config t

Router(config)#hostname R2

R2(config)#int Fa0/0

R2(config-if)#ip add 192.168.3.1 255.255.255.0

R2(config-if)#no shut

R2(config-if)#exit

R2(config)#int Se0/0/0

R2(config-if)#ip add 192.168.2.2 255.255.255.0

R2(config-if)#no shut

R2(config-if)#exit

R2(config)#int Se0/0/1

R2(config-if)#ip add 192.168.4.2 255.255.255.0

R2(config-if)#no shut

R2(config-if)#exit

R2(config)#ip route 0.0.0.0 0.0.0.0 192.168.2.1

R2(config)#ip route 0.0.0.0 0.0.0.0 192.168.4.1

R2(config)#router rip

R2(config-router)#network 192.168.2.0

R2(config-router)#network 192.168.3.0

R2(config-router)#network 192.168.4.0

R2(config-router)#end

R2#show ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is 192.168.4.1 to network 0.0.0.0

R 192.168.1.0/24 [120/1] via 192.168.2.1, 00:00:04, Serial0/0/0

C 192.168.2.0/24 is directly connected, Serial0/0/0

C 192.168.3.0/24 is directly connected, FastEthernet0/0

C 192.168.4.0/24 is directly connected, Serial0/0/1

R 192.168.5.0/24 [120/1] via 192.168.4.1, 00:00:15, Serial0/0/1

S* 0.0.0.0/0 [1/0] via 192.168.4.1

[1/0] via 192.168.2.1

R2#show ip protocols

Routing Protocol is "rip"

Sending updates every 30 seconds, next due in 28 seconds

Invalid after 180 seconds, hold down 180, flushed after 240

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Redistributing: rip

Default version control: send version 1, receive any version

Interface Send Recv Triggered RIP Key-chain

FastEthernet0/0 12 1

Serial0/0/1 12 1

Serial0/0/0 12 1

Automatic network summarization is in effect

Maximum path: 4

Routing for Networks:

192.168.2.0

192.168.3.0

192.168.4.0

Passive Interface(s):

Routing Information Sources:

Gateway Distance Last Update

192.168.2.1 120 00:00:10

192.168.4.1 120 00:00:21

Distance: (default is 120)

R2#debug ip rip

RIP protocol debugging is on

R2#RIP: received v1 update from 192.168.2.1 on Serial0/0/0

192.168.1.0 in 1 hops

RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (192.168.3.1)

RIP: build update entries

network 192.168.1.0 metric 2

network 192.168.2.0 metric 1

network 192.168.4.0 metric 1

network 192.168.5.0 metric 2

RIP: sending v1 update to 255.255.255.255 via Serial0/0/1 (192.168.4.2)

RIP: build update entries

network 192.168.1.0 metric 2

network 192.168.2.0 metric 1

network 192.168.3.0 metric 1

RIP: sending v1 update to 255.255.255.255 via Serial0/0/0 (192.168.2.2)

RIP: build update entries

network 192.168.3.0 metric 1

network 192.168.4.0 metric 1

network 192.168.5.0 metric 2

RIP: received v1 update from 192.168.4.1 on Serial0/0/1

192.168.5.0 in 1 hops

RIP: received v1 update from 192.168.2.1 on Serial0/0/0

192.168.1.0 in 1 hops

RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (192.168.3.1)

RIP: build update entries

network 192.168.1.0 metric 2

network 192.168.2.0 metric 1

network 192.168.4.0 metric 1

network 192.168.5.0 metric 2

RIP: sending v1 update to 255.255.255.255 via Serial0/0/1 (192.168.4.2)

RIP: build update entries

network 192.168.1.0 metric 2

network 192.168.2.0 metric 1

network 192.168.3.0 metric 1

RIP: sending v1 update to 255.255.255.255 via Serial0/0/0 (192.168.2.2)

RIP: build update entries

network 192.168.3.0 metric 1

network 192.168.4.0 metric 1

network 192.168.5.0 metric 2

RIP: received v1 update from 192.168.4.1 on Serial0/0/1

192.168.5.0 in 1 hops

RIP: received v1 update from 192.168.2.1 on Serial0/0/0

192.168.1.0 in 1 hops

RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (192.168.3.1)

RIP: build update entries

network 192.168.1.0 metric 2

network 192.168.2.0 metric 1

network 192.168.4.0 metric 1

network 192.168.5.0 metric 2

RIP: sending v1 update to 255.255.255.255 via Serial0/0/1 (192.168.4.2)

RIP: build update entries

network 192.168.1.0 metric 2

network 192.168.2.0 metric 1

network 192.168.3.0 metric 1

RIP: sending v1 update to 255.255.255.255 via Serial0/0/0 (192.168.2.2)

RIP: build update entries

network 192.168.3.0 metric 1

network 192.168.4.0 metric 1

network 192.168.5.0 metric 2

For R3

Router>en

Router#config t

Router(config)#hostname R3

R3(config)#int Fa0/0

R3(config-if)#ip add 192.168.5.1 255.255.255.0

R3(config-if)#no shut

R3(config-if)#exit

R3(config)#int Se0/0/1

R3(config-if)#ip add 192.168.4.1 255.255.255.0

R3(config-if)#no shut

R3(config-if)#exit

R3(config)#ip route 0.0.0.0 0.0.0.0 192.168.4.2

R1(config)#router rip

R1(config-router)#network 192.168.4.0

R1(config-router)#network 192.168.5.0

R1(config-router)#end

R3#show ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is 192.168.4.2 to network 0.0.0.0

R 192.168.1.0/24 [120/2] via 192.168.4.2, 00:00:11, Serial0/0/1

R 192.168.2.0/24 [120/1] via 192.168.4.2, 00:00:11, Serial0/0/1

R 192.168.3.0/24 [120/1] via 192.168.4.2, 00:00:11, Serial0/0/1

C 192.168.4.0/24 is directly connected, Serial0/0/1

C 192.168.5.0/24 is directly connected, FastEthernet0/0

S* 0.0.0.0/0 [1/0] via 192.168.4.2

R3#show ip protocols

Routing Protocol is "rip"

Sending updates every 30 seconds, next due in 15 seconds

Invalid after 180 seconds, hold down 180, flushed after 240

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Redistributing: rip

Default version control: send version 1, receive any version

Interface Send Recv Triggered RIP Key-chain

FastEthernet0/0 12 1

Serial0/0/1 12 1

Automatic network summarization is in effect

Maximum path: 4

Routing for Networks:

192.168.4.0

192.168.5.0

Passive Interface(s):

Routing Information Sources:

Gateway Distance Last Update

192.168.4.2 120 00:00:17

Distance: (default is 120)

R3#debug ip rip

RIP protocol debugging is on

R3#RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (192.168.5.1)

RIP: build update entries

network 192.168.1.0 metric 3

network 192.168.2.0 metric 2

network 192.168.3.0 metric 2

network 192.168.4.0 metric 1

RIP: sending v1 update to 255.255.255.255 via Serial0/0/1 (192.168.4.1)

RIP: build update entries

network 192.168.5.0 metric 1

RIP: received v1 update from 192.168.4.2 on Serial0/0/1

192.168.1.0 in 2 hops

192.168.2.0 in 1 hops

192.168.3.0 in 1 hops

RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (192.168.5.1)

RIP: build update entries

network 192.168.1.0 metric 3

network 192.168.2.0 metric 2

network 192.168.3.0 metric 2

network 192.168.4.0 metric 1

RIP: sending v1 update to 255.255.255.255 via Serial0/0/1 (192.168.4.1)

RIP: build update entries

network 192.168.5.0 metric 1

RIP: received v1 update from 192.168.4.2 on Serial0/0/1

192.168.1.0 in 2 hops

192.168.2.0 in 1 hops

192.168.3.0 in 1 hops

RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (192.168.5.1)

RIP: build update entries

network 192.168.1.0 metric 3

network 192.168.2.0 metric 2

network 192.168.3.0 metric 2

network 192.168.4.0 metric 1

RIP: sending v1 update to 255.255.255.255 via Serial0/0/1 (192.168.4.1)

RIP: build update entries

network 192.168.5.0 metric 1

RIP: received v1 update from 192.168.4.2 on Serial0/0/1

192.168.1.0 in 2 hops

192.168.2.0 in 1 hops

192.168.3.0 in 1 hops

RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (192.168.5.1)

RIP: build update entries

network 192.168.1.0 metric 3

network 192.168.2.0 metric 2

network 192.168.3.0 metric 2

network 192.168.4.0 metric 1

RIP: sending v1 update to 255.255.255.255 via Serial0/0/1 (192.168.4.1)

RIP: build update entries

network 192.168.5.0 metric 1

RIP: received v1 update from 192.168.4.2 on Serial0/0/1

192.168.1.0 in 2 hops

192.168.2.0 in 1 hops

192.168.3.0 in 1 hops

RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (192.168.5.1)

RIP: build update entries

network 192.168.1.0 metric 3

network 192.168.2.0 metric 2

network 192.168.3.0 metric 2

network 192.168.4.0 metric 1

RIP: sending v1 update to 255.255.255.255 via Serial0/0/1 (192.168.4.1)

RIP: build update entries

network 192.168.5.0 metric 1

RIP: received v1 update from 192.168.4.2 on Serial0/0/1

192.168.1.0 in 2 hops

192.168.2.0 in 1 hops

192.168.3.0 in 1 hops

RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (192.168.5.1)

RIP: build update entries

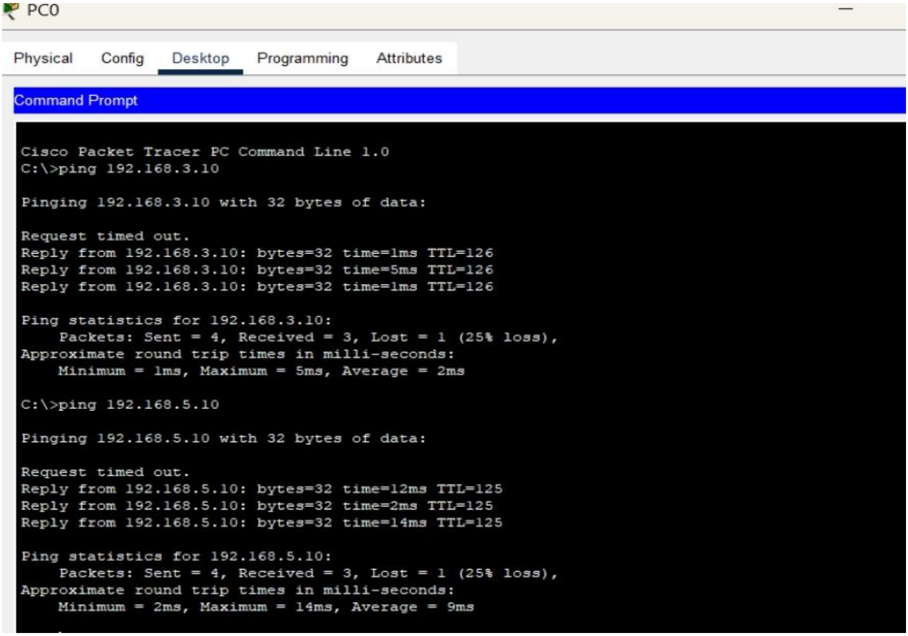
network 192.168.1.0 metric 3

network 192.168.2.0 metric 2

network 192.168.3.0 metric 2

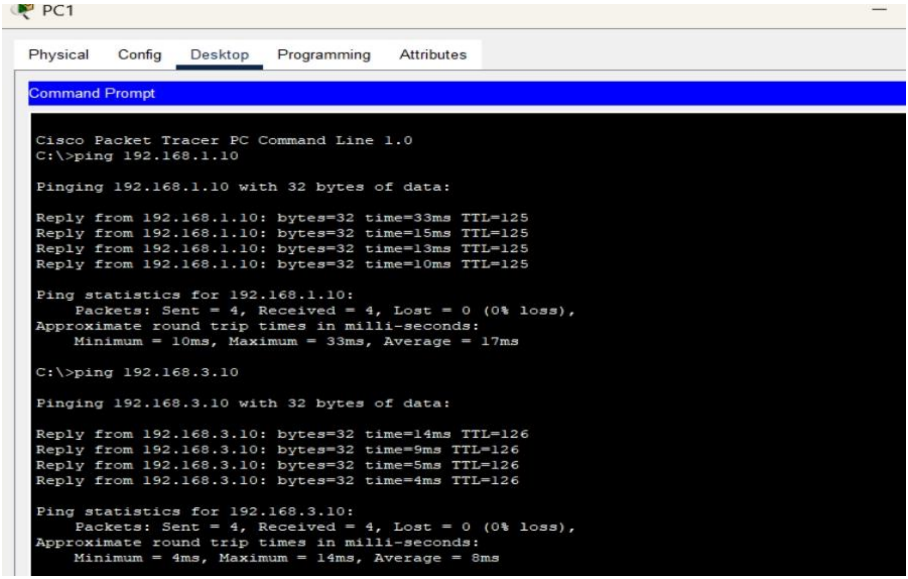
network 192.168.4.0 metric 1

For PC0




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.10: bytes=32 time=1ms TTL=126
Reply from 192.168.3.10: bytes=32 time=5ms TTL=126
Reply from 192.168.3.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 5ms, Average = 2ms

C:\>ping 192.168.5.10

Pinging 192.168.5.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.5.10: bytes=32 time=12ms TTL=125
Reply from 192.168.5.10: bytes=32 time=2ms TTL=125
Reply from 192.168.5.10: bytes=32 time=14ms TTL=125

Ping statistics for 192.168.5.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 14ms, Average = 9ms
```

For PC1




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=33ms TTL=125
Reply from 192.168.1.10: bytes=32 time=15ms TTL=125
Reply from 192.168.1.10: bytes=32 time=13ms TTL=125
Reply from 192.168.1.10: bytes=32 time=10ms TTL=125

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 33ms, Average = 17ms

C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.3.10: bytes=32 time=14ms TTL=126
Reply from 192.168.3.10: bytes=32 time=9ms TTL=126
Reply from 192.168.3.10: bytes=32 time=5ms TTL=126
Reply from 192.168.3.10: bytes=32 time=4ms TTL=126

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 14ms, Average = 8ms
```

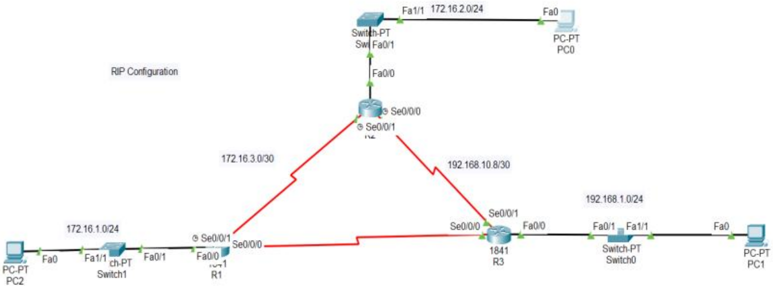
For PC2

```
C:\>ping 192.168.44.5

Pinging 192.168.44.5 with 32 bytes of data:

Reply from 192.168.44.5: bytes=32 time<1ms TTL=128
Reply from 192.168.44.5: bytes=32 time<1ms TTL=128
Reply from 192.168.44.5: bytes=32 time<1ms TTL=128
Reply from 192.168.44.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.44.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.20.123.5

Pinging 200.20.123.5 with 32 bytes of data:

Reply from 200.20.123.5: bytes=32 time<1ms TTL=126
Reply from 200.20.123.5: bytes=32 time=11ms TTL=126
Reply from 200.20.123.5: bytes=32 time=11ms TTL=126
Reply from 200.20.123.5: bytes=32 time=12ms TTL=126

Ping statistics for 200.20.123.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 8ms
```

```
C:\>ping 192.168.44.5

Pinging 192.168.44.5 with 32 bytes of data:

Reply from 177.77.77.2: bytes=32 time<1ms TTL=126
Reply from 177.77.77.2: bytes=32 time=12ms TTL=126
Reply from 177.77.77.2: bytes=32 time=10ms TTL=126
Reply from 177.77.77.2: bytes=32 time=10ms TTL=126

Ping statistics for 192.168.44.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 8ms

C:\>ping 192.168.44.6

Pinging 192.168.44.6 with 32 bytes of data:

Reply from 177.77.77.1: bytes=32 time<1ms TTL=126
Reply from 177.77.77.1: bytes=32 time=14ms TTL=126
Reply from 177.77.77.1: bytes=32 time=15ms TTL=126
Reply from 177.77.77.1: bytes=32 time=13ms TTL=126

Ping statistics for 192.168.44.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 15ms, Average = 10ms

C:\>
```

Jobsheet-9

RIP Configuration



<u>FOR R1</u>

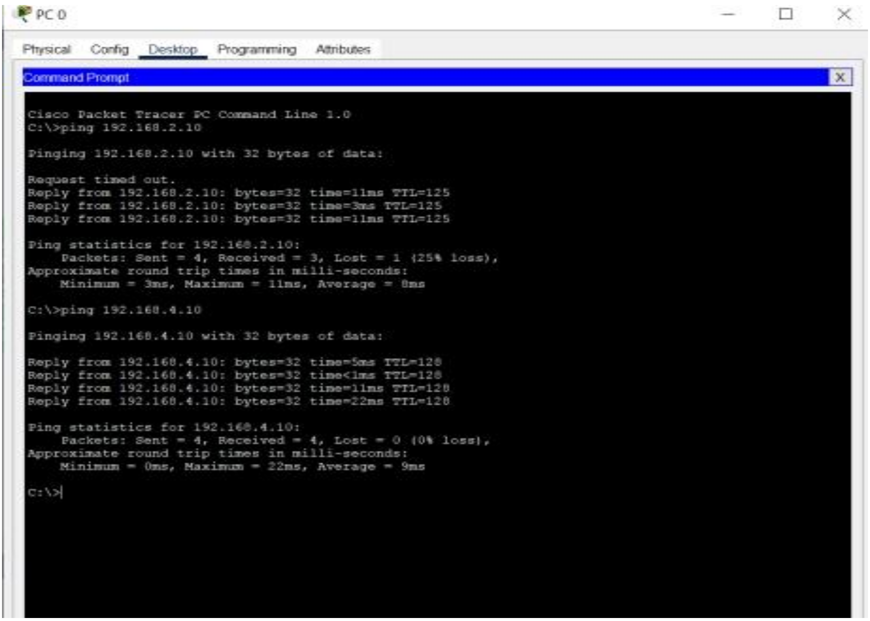
- R1(config)#router eigrp 1
- R1(config-router)#network 172.16.0.0
- R1(config-router)#network 192.168.10.4 0.0.0.3
- R1(config-router)#end

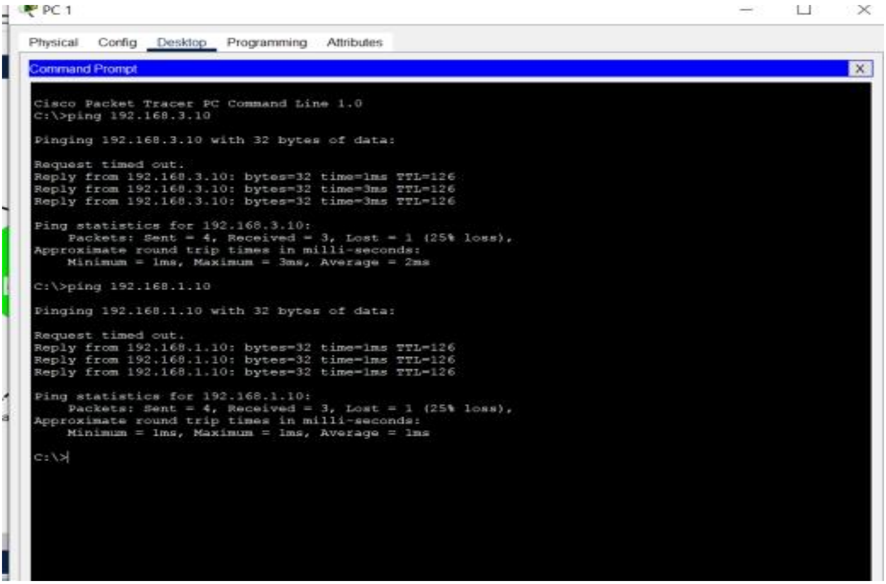
<u>FOR R2</u>

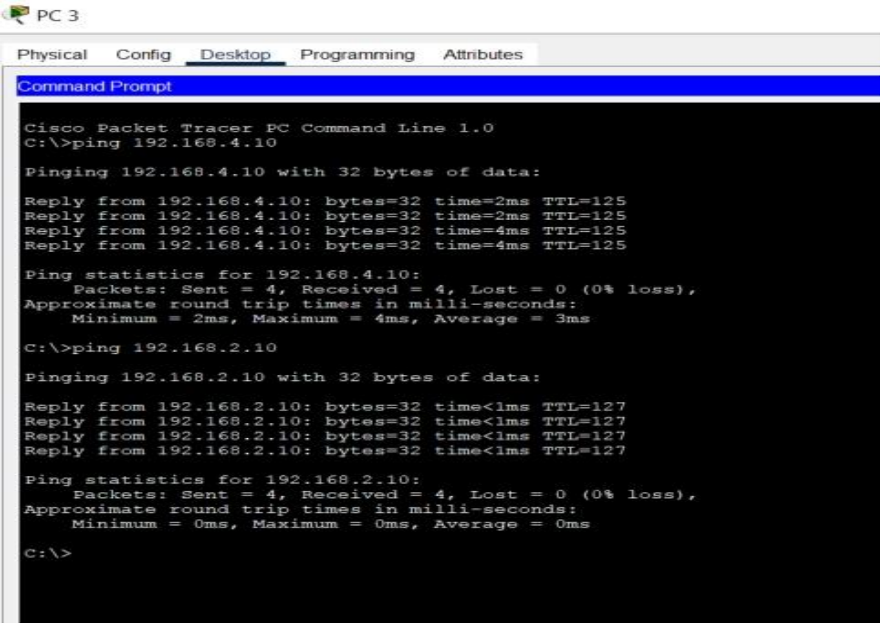
- R2(config)#route eigrp 1
- R2(config-router)#network 172.16.0.0
- R2(config-router)#network 192.168.10.8 0.0.0.3
- R2(config-router)#end

FOR R3

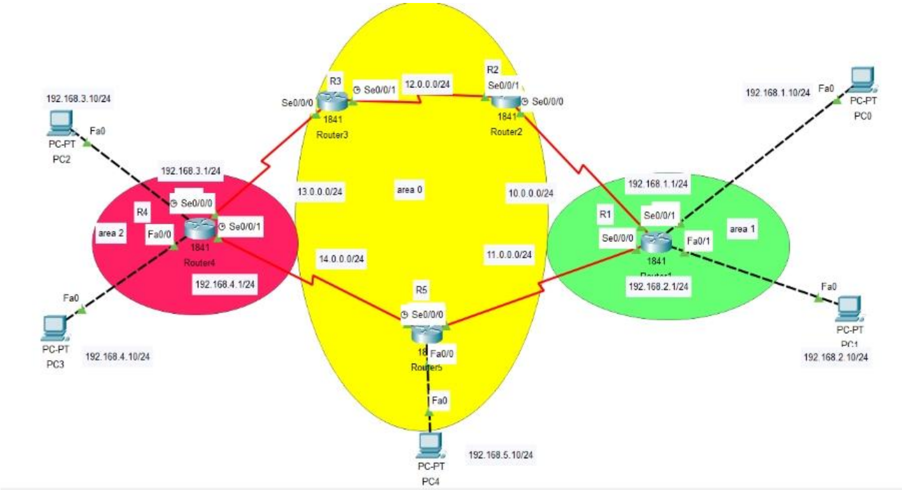
- R3(config)# router eigrp 1
- R3(config-router)#network 192.168.1.0 0.0.0.255
- R3(config-router)#network 192.168.10.4 0.0.0.3
- R3(config-router)#network 192.168.10.8 0.0.0.3
- R3(config-router)#end




PC 1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.10
Pinging 192.168.3.10 with 32 bytes of data:
Request timed out.
Reply from 192.168.3.10: bytes=32 time=1ms TTL=126
Reply from 192.168.3.10: bytes=32 time=3ms TTL=126
Reply from 192.168.3.10: bytes=32 time=3ms TTL=126
Ping statistics for 192.168.3.10:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 3ms, Average = 2ms
C:\>ping 192.168.1.10
Pinging 192.168.1.10 with 32 bytes of data:
Request timed out.
Reply from 192.168.1.10: bytes=32 time=1ms TTL=126
Reply from 192.168.1.10: bytes=32 time=1ms TTL=126
Reply from 192.168.1.10: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.1.10:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>


PC 3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.4.10
Pinging 192.168.4.10 with 32 bytes of data:
Reply from 192.168.4.10: bytes=32 time=2ms TTL=125
Reply from 192.168.4.10: bytes=32 time=2ms TTL=125
Reply from 192.168.4.10: bytes=32 time=4ms TTL=125
Reply from 192.168.4.10: bytes=32 time=4ms TTL=125
Ping statistics for 192.168.4.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 4ms, Average = 3ms
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.2.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>

Jobsheet10



Devices	Ip address	Subnet Mask	Gateway
PC0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	192.168.2.10	255.255.255.0	192.168.2.1
PC2	192.168.3.10	255.255.255.0	192.168.3.1
PC3	192.168.4.10	255.255.255.0	192.168.4.1
PC4	192.168.5.10	255.255.255.0	192.168.5.1

For router1

Router>en

Router#config t

Router(config)#hostname R1

R1(config)#int Fa0/0

R1(config-if)#ip add 192.168.1.1 255.255.255.0

R1(config-if)#no shut

R1(config-if)#exit

R1(config)#int Fa0/1

R1(config-if)#ip add 192.168.2.1 255.255.255.0

R1(config-if)#no shut

R1(config-if)#exit

R1(config)#int Se0/0/1

R1(config-if)#ip add 10.0.0.1 255.255.255.0

R1(config-if)#no shut

R1(config-if)#exit

```
R1(config)#int Se0/0/0

R1(config-if)#ip add 11.0.0.1 255.255.255.0

R1(config-if)#no shut

R1(config-if)#exit


R1(config)#router ospf 1

R1(config-router)#router-id 10.0.0.1

R1(config-router)#log-adjacency-changes

R1(config-router)#network 10.0.0.0 0.0.0.255 area 1

R1(config-router)#network 11.0.0.0 0.0.0.255 area1

R1(config-router)# network 192.168.1.0 0.0.0.255 area1

R1(config-router)# network 192.168.2.0 0.0.0.255 area1

R1(config-router)#end


R1#show ip protocols
```

Routing Protocol is "ospf 1"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 10.0.0.1

Number of areas in this router is 1. 1 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

10.0.0.0 0.0.0.255 area 1

11.0.0.0 0.0.0.255 area 1

192.168.1.0 0.0.0.255 area 1

192.168.2.0 0.0.0.255 area 1

Routing Information Sources:

Gateway Distance Last Update

10.0.0.1 110 00:00:36

12.0.0.2 110 00:00:36

14.0.0.2 110 00:00:36

Distance: (default is 110)

For R2

Router>en

Router#config t

Router(config)#hostname R2

R2(config)#int Se0/0/1

R2(config-if)#ip add 12.0.0.2 255.255.255.0

R2(config-if)#no shut

R2(config-if)#exit

R2(config)#int Se0/0/0

R2(config-if)#ip add 10.0.0.2 255.255.255.0

R2(config-if)#no shut

```
R2(config-if)#exit


R2(config)#router ospf 1

R2(config-router)#router-id 12.0.0.2

R2(config-router)#log-adjacency-changes

R2(config-router)#network 12.0.0.0 0.0.0.255 area 0

R2(config-router)#network 10.0.0.0 0.0.0.255 area1

R2(config-router)#end


R2#show ip protocols


Routing Protocol is "ospf 1"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 12.0.0.2
```

Number of areas in this router is 2. 2 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

12.0.0.0 0.0.0.255 area 0

10.0.0.0 0.0.0.255 area 1

Routing Information Sources:

Gateway Distance Last Update

10.0.0.1 110 00:01:20

12.0.0.2 110 00:01:20

13.0.0.2 110 00:01:19

14.0.0.2 110 00:01:20

Distance: (default is 110)

For R3

Router>en

```
Router#config t

Router(config)#hostname R3

R3(config)#int Se0/0/1

R3(config-if)#ip add 12.0.0.1 255.255.255.0

R3(config-if)#no shut

R3(config-if)#exit

R3(config)#int Se0/0/0

R3(config-if)#ip add 13.0.0.2 255.255.255.0

R3(config-if)#no shut

R3(config-if)#exit


R3(config)#router ospf 1

R3(config-router)#router-id 13.0.0.2

R3(config-router)#log-adjacency-changes

R3(config-router)#network 13.0.0.0 0.0.0.255 area 2
```

R3(config-router)#network 12.0.0.0 0.0.0.255 area 0

R3(config-router)#end

R3#show ip protocols

Routing Protocol is "ospf 1"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 13.0.0.2

Number of areas in this router is 2. 2 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

13.0.0.0 0.0.0.255 area 2

12.0.0.0 0.0.0.255 area 0

Routing Information Sources:

Gateway Distance Last Update

12.0.0.2 110 00:02:00

13.0.0.1 110 00:01:56

13.0.0.2 110 00:02:00

14.0.0.2 110 00:01:56

Distance: (default is 110)

For R4

Router>en

Router#config t

Router(config)#hostname R4

R4(config)#int Fa0/0

R4(config-if)#ip add 192.168.4.1 255.255.255.0

R4(config-if)#no shut

R4(config-if)#exit

R4(config)#int Fa0/1

R4(config-if)#ip add 192.168.3.1 255.255.255.0

R4(config-if)#no shut

R4(config-if)#exit

R4(config)#int Se0/0/1

R4(config-if)#ip add 14.0.0.1 255.255.255.0

R4(config-if)#no shut

R4(config-if)#exit

R4(config)#int Se0/0/0

R4(config-if)#ip add 13.0.0.1 255.255.255.0

R4(config-if)#no shut

R4(config-if)#exit

R4(config)#router ospf 1

R4(config-router)#router-id 13.0.0.1

R4(config-router)#log-adjacency-changes

R4(config-router)#network 13.0.0.0 0.0.0.255 area 2

R4(config-router)#network 14.0.0.0 0.0.0.255 area 2

R4(config-router)#network 192.168.3.0 0.0.0.255 area 2

R4(config-router)#network 192.168.4.0 0.0.0.255 area 2

R4(config-router)#end

R4#show ip protocols

Routing Protocol is "ospf 1"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 13.0.0.1

Number of areas in this router is 1. 1 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

13.0.0.0 0.0.0.255 area 2

14.0.0.0 0.0.0.255 area 2

192.168.3.0 0.0.0.255 area 2

192.168.4.0 0.0.0.255 area 2

Routing Information Sources:

Gateway Distance Last Update

13.0.0.1 110 00:02:35

13.0.0.2 110 00:02:36

14.0.0.2 110 00:02:35

Distance: (default is 110)

For R5

```
Router>en

Router#config t

Router(config)#hostname R5

R5(config)#int Fa0/0

R5(config-if)#ip add 192.168.5.1 255.255.255.0

R5(config-if)#no shut

R5(config-if)#exit

R5(config)#int Se0/0/1

R5(config-if)#ip add 14.0.0.2 255.255.255.0

R5(config-if)#no shut

R5(config-if)#exit

R5(config)#int Se0/0/0

R5(config-if)#ip add 11.0.0.2 255.255.255.0

R5(config-if)#no shut

R5(config-if)#exit
```

R5(config)#router ospf 1

R5(config-router)#router-id 14.0.0.2

R5(config-router)#log-adjacency-changes

R5(config-router)#network 14.0.0.0 0.0.0.255 area 2

R5(config-router)#network 192.168.5.0 0.0.0.255 area 0

R5(config-router)#network 11.0.0.0 0.0.0.255 area 1

R5(config-router)#end

R5#show ip protocols

Routing Protocol is "ospf 1"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 14.0.0.2

Number of areas in this router is 3. 3 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

14.0.0.0 0.0.0.255 area 2

192.168.5.0 0.0.0.255 area 0

11.0.0.0 0.0.0.255 area 1

Routing Information Sources:

Gateway Distance Last Update

10.0.0.1 110 00:03:27

12.0.0.2 110 00:03:27

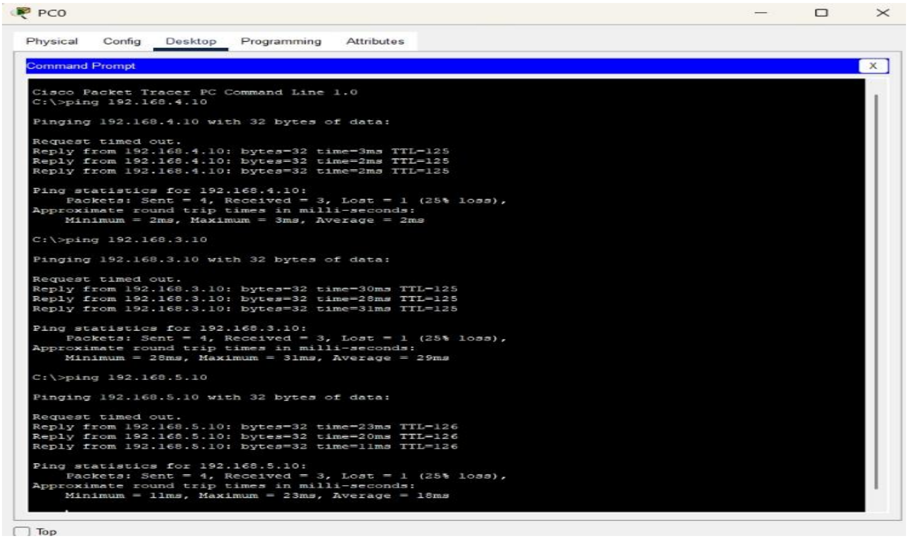
13.0.0.1 110 00:03:26

13.0.0.2 110 00:03:27

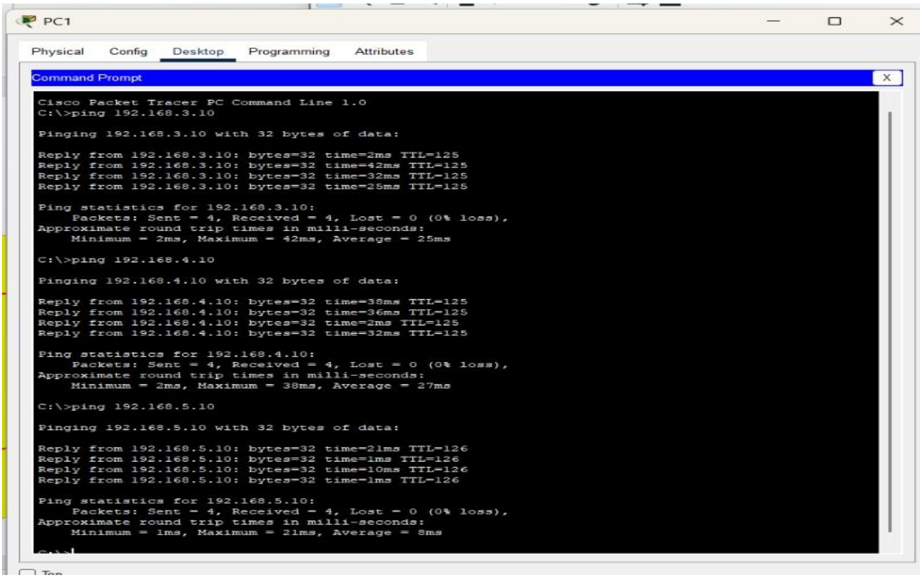
14.0.0.2 110 00:03:27

Distance: (default is 110)

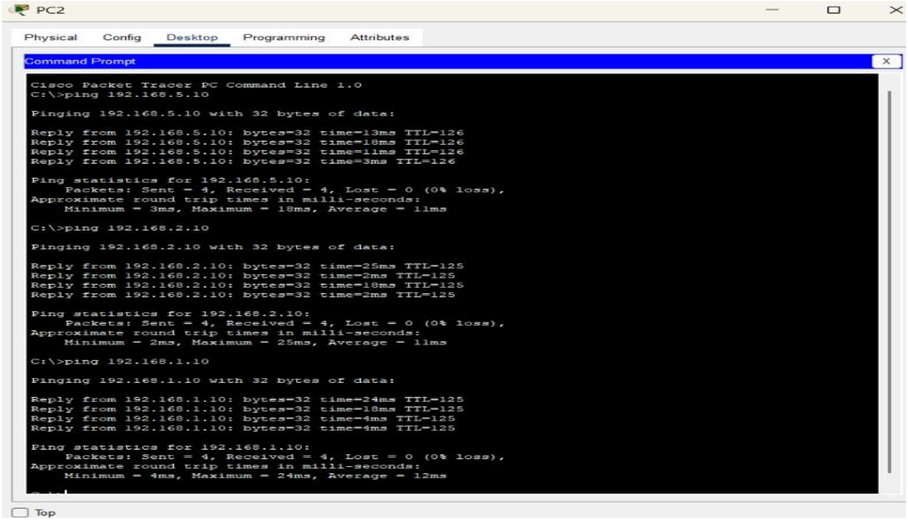
For PC0



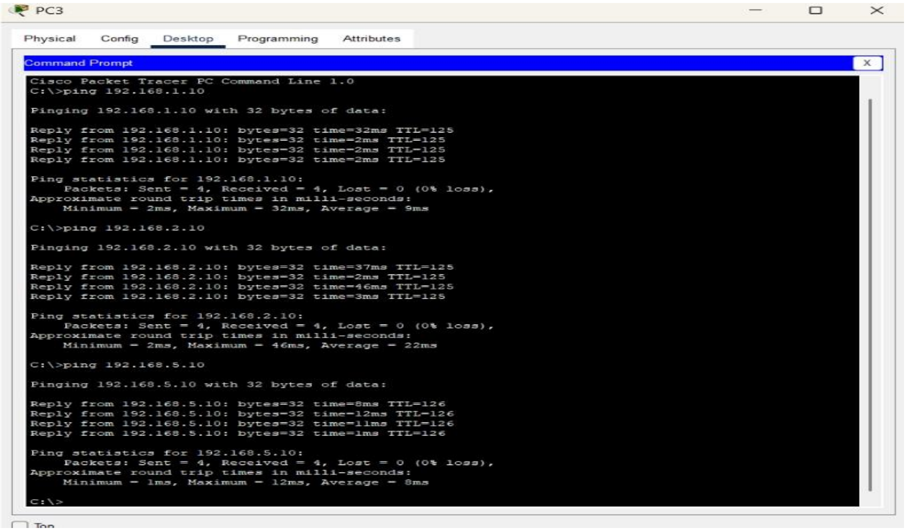
For PC1



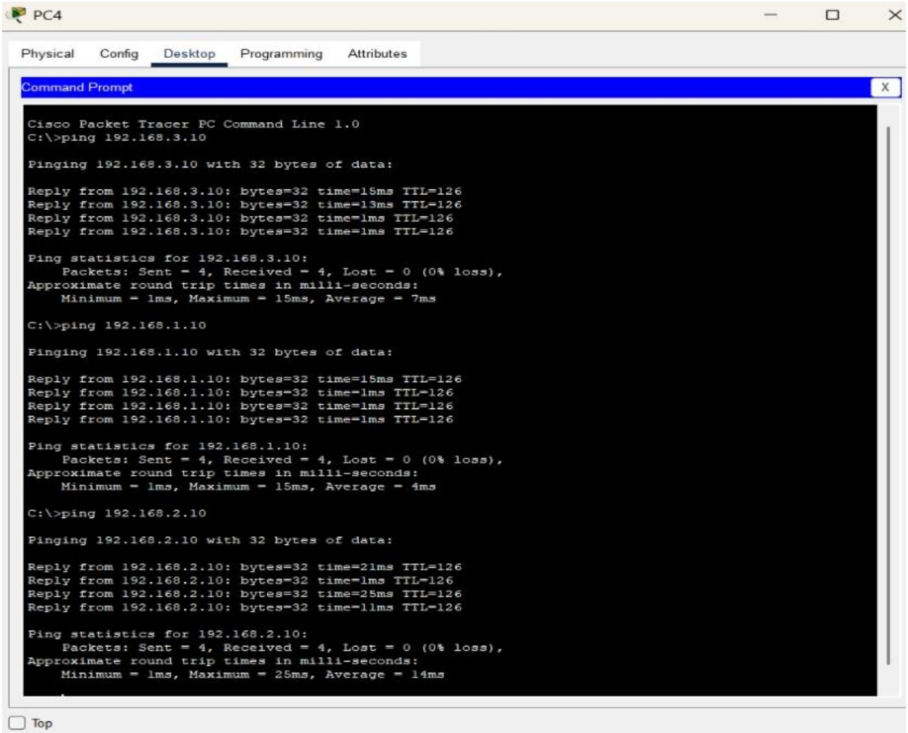
For PC2



For Pc3



For PC4


PC4
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.10
Pinging 192.168.3.10 with 32 bytes of data:
Reply from 192.168.3.10: bytes=32 time=15ms TTL=126
Reply from 192.168.3.10: bytes=32 time=13ms TTL=126
Reply from 192.168.3.10: bytes=32 time=1ms TTL=126
Reply from 192.168.3.10: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.3.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 15ms, Average = 7ms
C:\>ping 192.168.1.10
Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time=15ms TTL=126
Reply from 192.168.1.10: bytes=32 time=1ms TTL=126
Reply from 192.168.1.10: bytes=32 time=1ms TTL=126
Reply from 192.168.1.10: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.1.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 15ms, Average = 4ms
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Reply from 192.168.2.10: bytes=32 time=21ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=25ms TTL=126
Reply from 192.168.2.10: bytes=32 time=11ms TTL=126
Ping statistics for 192.168.2.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 25ms, Average = 14ms
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